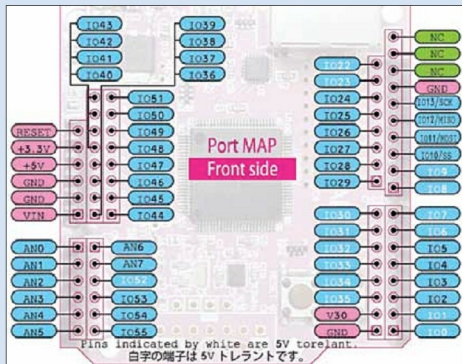


GR-Sakura: Technical Review by EFY

The Hardware of GR-Sakura board is very well organised to make accessible all the resources of 96MHz, 32-bit MCU while still maintaining compatibility with Arduino platform.

The CISC MCU used is from Renesas RX series, which is known for low power (500µA/MHz) and better code efficiency. This is a small but a very powerful chip with whole lot of communication functions that make networking very comfortable. The communication functions come in-built with the chip are Ethernet controller, two-channel USB, three-channel CAN, four-channel I²C and 13-channel serial communication, which are made accessible to play within GR-Sakura board.

Another great strength of the board is a bunch of 55 digital I/Os ready to be interfaced with the external world. So you can implement as many functions as you would like without falling short of the I/O pins.



I/O map of GR-Sakura

There are several other features of this board that are mentioned below, which makes it a complete development environment for hobbyists and students:

Xbee Interface. For wireless network, there are dedicated pins on the board to directly interface Xbee module. The module sticks at the back of the board and is instantly ready. **JTAG.** The JTAG pins allows on-chip debugging using E1 Emulator.

microSD Interface. This interface makes possible the exchange of data between the board and the PC.

The board is dual-powered (USB and DC adaptor). DC adaptor is used whenever driving loads more than the capacity of a USB port of the computer. The overall power supply design as seen from the schematic is reasonable enough for any kind of experimentation. Unlike Arduino, the adaptor here should be 5V instead of 9V.

Most of all, the Arduino compatible online compiler makes development very fast, and the best part is that the board can be programmed just by a drag and drop operation with no special skills required. These things open the doors for those people who may not have an electronics degree but have interest in electronics.

What is GR-Sakura?

‘GR’ prefix stands for ‘Gadget Renesas,’ which is a project that offers a series of boards bundled with an ‘easy to start with’ cloud environ-

ment for rapid prototyping. Like all open hardware boards, GR-Sakura is targeted at both beginners as well as engineers looking forward to do some rapid prototyping.

The board is powered by a power-

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